GPUs (architecture & *CUDA*)

21.04.2026

VSS Lab

CPU vs GPU?

Performance angle…

CPU >> GPU

?

GPU >> CPU

Performance angle…

CPU >> GPU	**For sequential problems**
GPU >> CPU	**For (embarrassingly) parallel problems**

Architecture (CPU vs GPU)

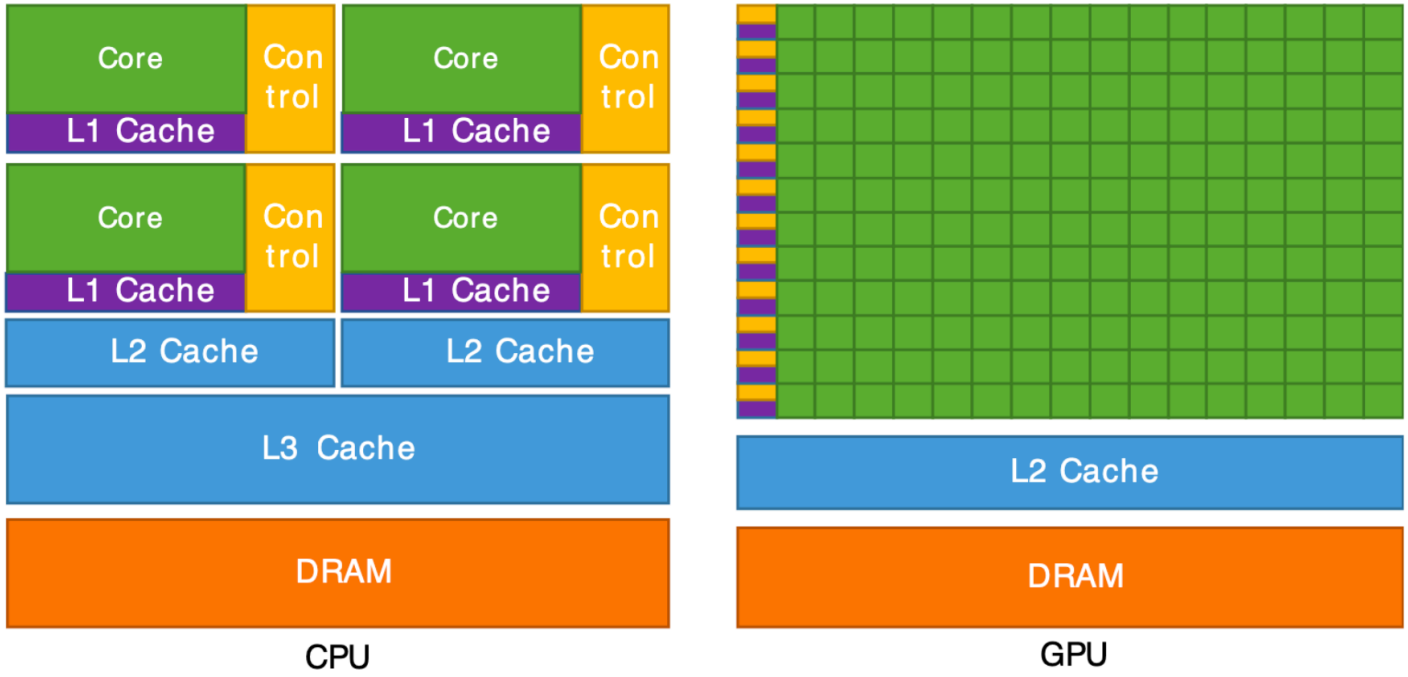
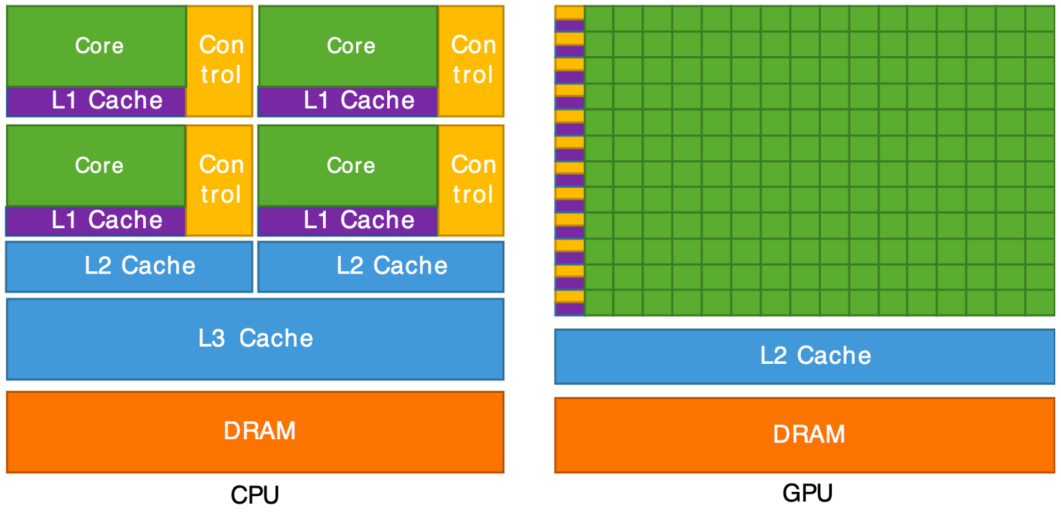


Image source: https://cvw.cac.cornell.edu/gpu-architecture/gpu-characteristics/design

Architecture (CPU vs GPU)

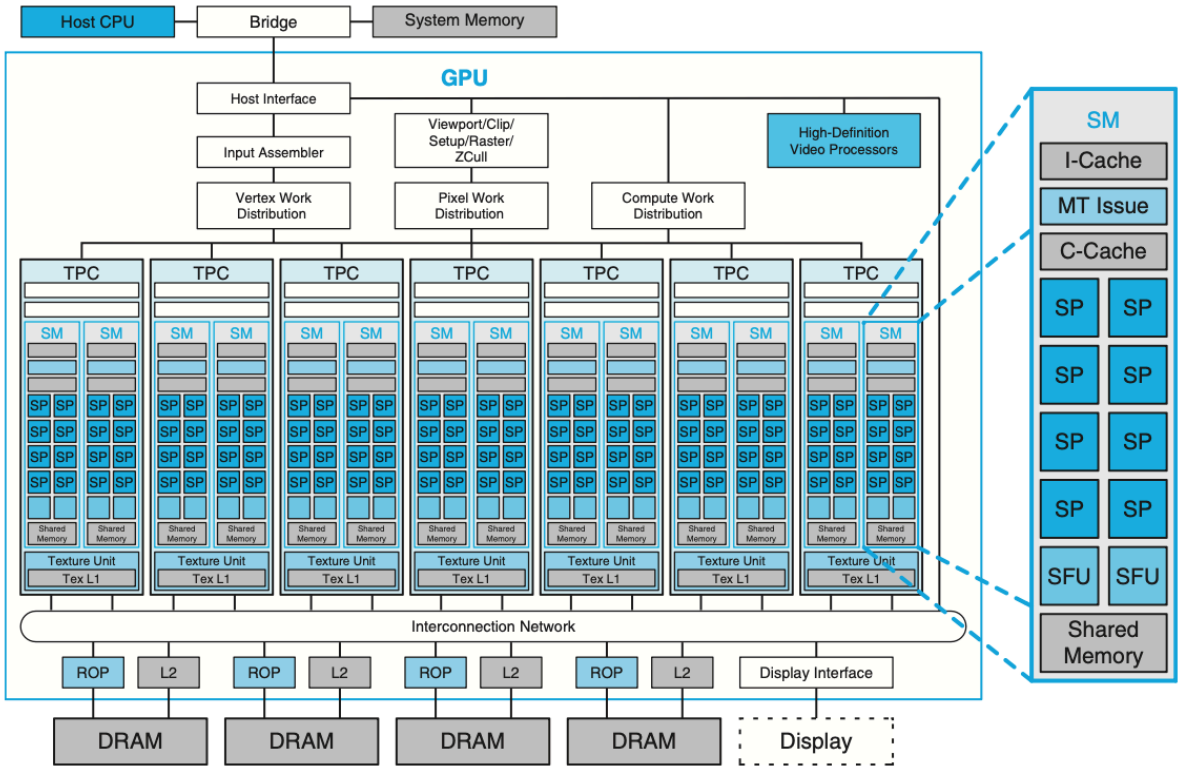


GPU >> CPU w.r.t. no. cores

GPUs have tens of thousands of threads >> No. CPU threads

Image source: https://cvw.cac.cornell.edu/gpu-architecture/gpu-characteristics/design

Architecture GPU (Tesla, 2006)



Focus on
SMs
SPs

Image source: https://www.amazon.com/Computer-Organization-Design-Fifth-Architecture/dp/B01MZ3QR8D

SM of a GPU today (2026)



Image source: https://developer.nvidia.com/blog/inside-nvidia-blackwell-ultra-the-chip-powering-the-ai-factory-era/


NVIDIA
CUDA

?

CUDA in context

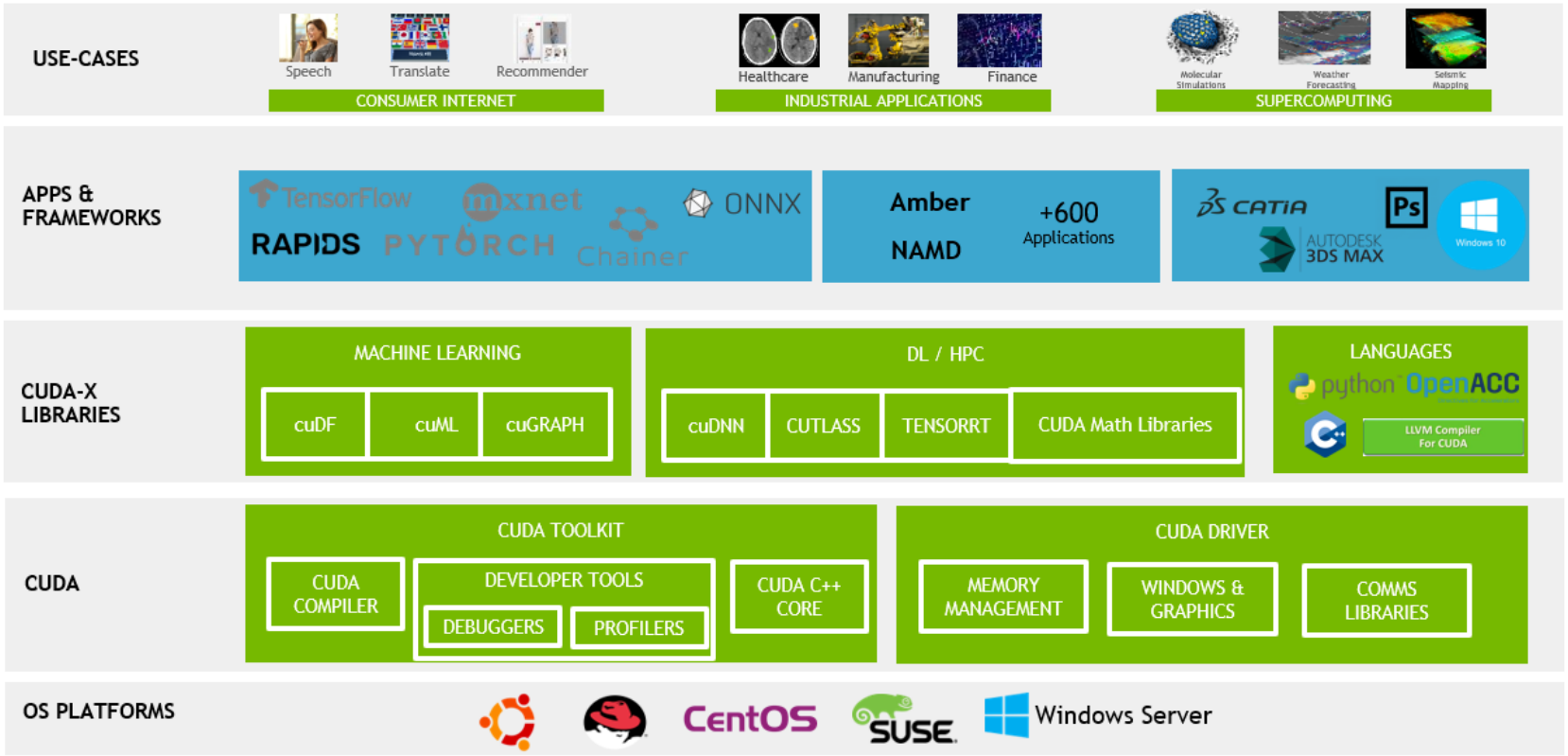


Image source: https://blogs.nvidia.com/blog/what-is-cuda-2/

What about actual execution on *HW*?

Execution on *HW*?

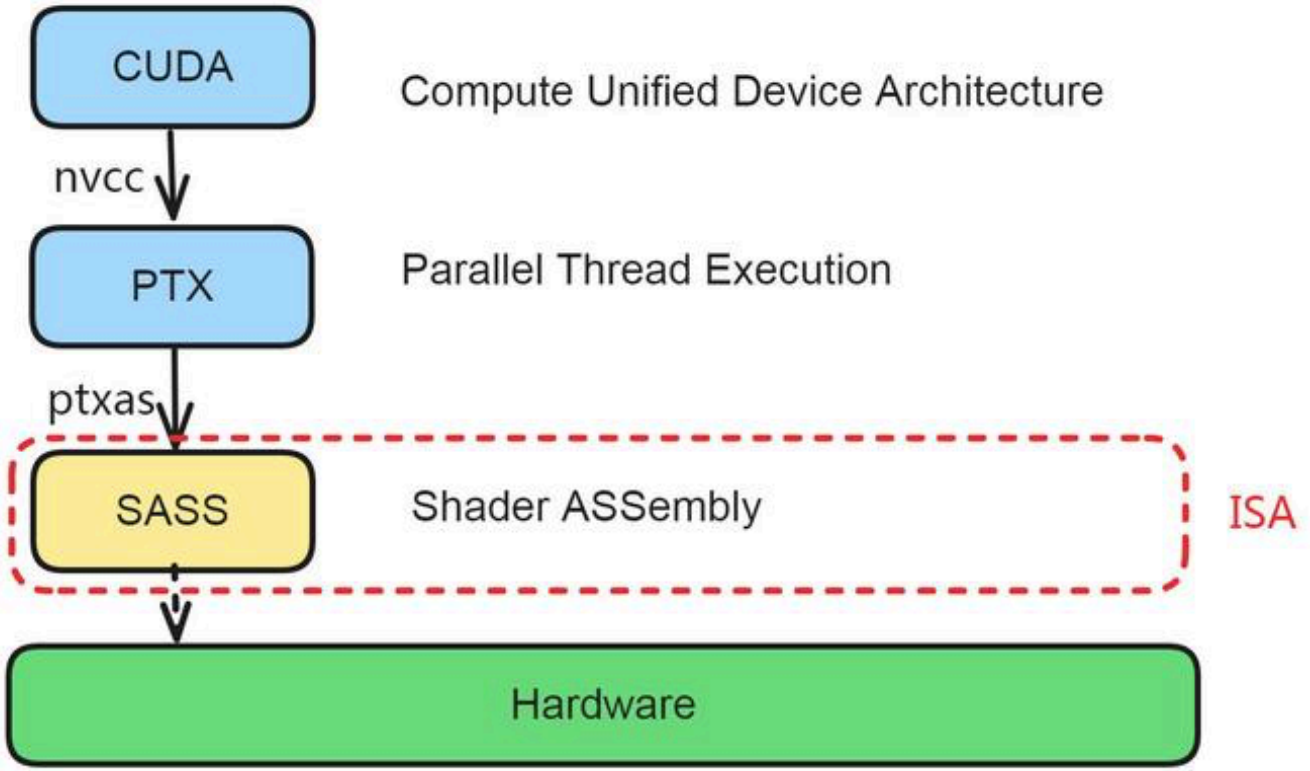


Image source: https://zhuanlan.zhihu.com/p/686198447

Hands-on exercise

Lab GPU: Architecture?